

Subject: Submission of Research Paper: Ranking football teams via metric-driven higher-order networks

To the Editors of Chaos, Solitons & Fractals,

I am pleased to submit our manuscript, "Ranking football teams via performance-driven higher-order networks", for consideration as a research article in Chaos, Solitons & Fractals.

In this work, we introduce a novel methodological framework that integrates HodgeRank with event-level performance metrics to evaluate competitive dynamics in professional football. Unlike traditional rating systems (such as Elo) that rely solely on match outcomes, our approach maps performance data onto weighted networks, using the gradient component of the Hodge decomposition to derive team rankings.

Key highlights of our study include:

- The development of a unified framework that captures higher-order interactions in sports data.
- A comparative analysis of the five major European leagues, revealing significant league-dependent variations in how performance metrics correlate with official standings.
- The optimization of a composite rating that identifies statistical discrepancies between on-field performance and accumulated points (the "luck factor").

Given the journal's focus on the applications of network science and nonlinear dynamics to real-world systems, we believe our findings regarding the structural inconsistencies and competitive patterns of these leagues will be of great interest to your readership.

This manuscript is original, has not been published elsewhere, and is not under consideration by another journal. All authors have approved the submission.

Thank you for your time and consideration.

Sincerely,

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